DEVOPS LABORATORY

Viva Voce Question Bank with Answers

50 Questions & Answers | 10 Experiments | B.Tech Laboratory

Instructions: Prepare the definitions, concepts, differences, and practical commands. Answers are kept concise for quick viva voce revision.

EXPERIMENT 1: DevOps – Principles, Practices, and DevOps Engineer Roles and Responsibilities

1. What is DevOps?

Answer: DevOps is a combination of Development (Dev) and Operations (Ops). It is a software development approach that improves collaboration, automation, and faster software delivery.

2. What are the main objectives of DevOps?

Answer: Faster software delivery, improved collaboration, Continuous Integration and Continuous Delivery, better software quality, and reduced deployment failures.

3. What are the main principles of DevOps?

Answer: Collaboration, automation, continuous integration, continuous delivery, continuous monitoring, and continuous improvement.

4. What is the role of a DevOps Engineer?

Answer: A DevOps Engineer manages development and deployment processes, automates workflows, maintains CI/CD pipelines, manages infrastructure, and monitors applications.

5. What is CI/CD in DevOps?

Answer: CI stands for Continuous Integration, where developers frequently integrate code changes. CD stands for Continuous Delivery or Continuous Deployment, which automates software release processes.

EXPERIMENT 2: Explore Version Control System Tools for Source Code Management

6. What is a Version Control System (VCS)?

Answer: A Version Control System is a tool used to track changes in source code and maintain different versions of files.

7. What are the types of Version Control Systems?

Answer: The three main types are Local Version Control Systems, Centralized Version Control Systems, and Distributed Version Control Systems.

8. What is Git?

Answer: Git is a distributed version control system used to track source code changes and support collaboration among developers.

9. What is the difference between centralized and distributed version control systems?

Answer: In a centralized system, code is stored on a central server. In a distributed system, every developer has a complete local copy of the repository and its history.

10. What are some popular Version Control System tools?

Answer: Git, SVN (Subversion), Mercurial, and Perforce.

EXPERIMENT 3: Install Git, Create a GitHub Account, and Execute Git Operations

11. What is GitHub?

Answer: GitHub is a cloud-based platform used to host Git repositories and collaborate with other developers.

12. What is a Git repository?

Answer: A Git repository is a storage location that maintains project files and their complete version history.

13. What is the purpose of the git init command?

Answer: The git init command initializes a new Git repository in the current directory.

14. What is the difference between git add and git commit?

Answer: git add moves changes to the staging area, whereas git commit records the staged changes in the repository history.

15. What is the purpose of git push and git pull?

Answer: git push uploads local commits to a remote repository. git pull downloads and integrates changes from a remote repository.

EXPERIMENT 4: Installing and Configuring Jenkins to Set Up a Build Job

16. What is Jenkins?

Answer: Jenkins is an open-source automation server used to automate software building, testing, and deployment processes.

17. What is a Jenkins Job?

Answer: A Jenkins Job is a configured task that performs operations such as building, testing, or deploying an application.

18. What is a Jenkins Pipeline?

Answer: A Jenkins Pipeline is a sequence of automated steps used to build, test, and deploy software.

19. What is a Jenkinsfile?

Answer: A Jenkinsfile is a text file that defines the Jenkins Pipeline using Declarative or Scripted Pipeline syntax.

20. What are the advantages of Jenkins?

Answer: Jenkins is open source, supports CI/CD, provides many plugins, integrates with Git and GitHub, and automates build and testing processes.

EXPERIMENT 5: Jenkins Master-Slave Architecture and Implementing Slave Nodes

21. What is Jenkins Master-Slave Architecture?

Answer: It is an architecture in which the Jenkins Controller manages jobs and schedules tasks, while Jenkins Agents execute those tasks.

22. What is a Jenkins Controller?

Answer: The Jenkins Controller is the central component that manages configurations, schedules jobs, and assigns tasks to agents.

23. What is a Jenkins Agent?

Answer: A Jenkins Agent is a machine or process that executes build and testing tasks assigned by the Jenkins Controller.

24. Why do we use Jenkins Agents?

Answer: Jenkins Agents distribute workloads, support parallel builds, and improve the scalability of the Jenkins environment.

25. What is the difference between Jenkins Controller and Agent?

Answer: The Controller manages and schedules jobs, whereas the Agent executes the assigned jobs.

EXPERIMENT 6: Docker Architecture and Container Life Cycle

26. What is Docker?

Answer: Docker is an open-source platform used to develop, package, and run applications inside lightweight containers.

27. What is a Docker Image?

Answer: A Docker Image is a read-only template containing application code, libraries, dependencies, and configuration required to create a container.

28. What is a Docker Container?

Answer: A Docker Container is a running or stopped instance of a Docker Image that provides an isolated environment for an application.

29. What are the main components of Docker Architecture?

Answer: The main components are Docker Client, Docker Daemon, Docker Images, Docker Containers, and Docker Registry.

30. What are some important Docker commands?

Answer: docker --version checks the version; docker pull downloads an image; docker images lists images; docker run creates and starts a container; docker ps lists running containers; docker stop stops a container; docker rm removes a container; and docker rmi removes an image.

EXPERIMENT 7: Dockerfile Instructions and Building an Image for a Web Application

31. What is a Dockerfile?

Answer: A Dockerfile is a text file containing instructions to build a Docker Image automatically.

32. What are the important instructions used in a Dockerfile?

Answer: FROM specifies the base image; RUN executes commands during image building; COPY copies files; WORKDIR sets the working directory; EXPOSE documents a port; CMD specifies a default command; and ENTRYPOINT defines the main executable.

33. What is the purpose of the docker build command?

Answer: The docker build command creates a Docker Image using instructions provided in a Dockerfile.

34. What is the difference between CMD and ENTRYPOINT?

Answer: CMD provides default commands or arguments for a container. ENTRYPOINT defines the main executable that runs when the container starts.

35. What is Docker Hub?

Answer: Docker Hub is a cloud-based registry used to store, share, and download Docker Images.

EXPERIMENT 8: Deploy a Containerized Application on a Kubernetes Cluster

36. What is Kubernetes?

Answer: Kubernetes is an open-source container orchestration platform used to deploy, manage, scale, and monitor containerized applications.

37. What is a Kubernetes Cluster?

Answer: A Kubernetes Cluster is a collection of machines called nodes that work together to run containerized applications.

38. What is a Pod in Kubernetes?

Answer: A Pod is the smallest deployable unit in Kubernetes. It contains one or more containers that share network and storage resources.

39. What is a Kubernetes Deployment?

Answer: A Deployment manages application Pods and ensures the required number of replicas are running. It also supports rolling updates and rollbacks.

40. What is the purpose of the kubectl command?

Answer: kubectl is a command-line tool used to communicate with and manage a Kubernetes cluster.

EXPERIMENT 9: Install and Configure Pull-Based Software Configuration Management Using Puppet

41. What is Puppet?

Answer: Puppet is an open-source configuration management and automation tool used to manage infrastructure and maintain the desired system configuration.

42. What is Configuration Management?

Answer: Configuration Management is the process of maintaining systems in a consistent and desired state by managing software, configurations, and infrastructure.

43. What is Pull-Based Configuration Management?

Answer: In a pull-based model, agents periodically contact the central server to retrieve configuration information and apply the required changes.

44. What are Puppet Server and Puppet Agent?

Answer: The Puppet Server (traditionally called Puppet Master) manages configurations, while the Puppet Agent runs on client machines and retrieves configurations from the server.

45. What is a Puppet Manifest?

Answer: A Puppet Manifest is a file containing Puppet code that defines the desired configuration of a system. It generally uses the .pp extension.

EXPERIMENT 10: Software Configuration Management Using Puppet Building Blocks

46. What are the main building blocks of Puppet?

Answer: The main building blocks include Resources, Manifests, Classes, Modules, and Functions.

47. What is a Puppet Resource?

Answer: A Resource is the basic unit of Puppet configuration. It describes a system component such as a file, package, service, or user.

48. What is a Puppet Module?

Answer: A Puppet Module is a collection of manifests, files, templates, and other resources used to organize and reuse configuration code.

49. What is a Puppet Class?

Answer: A Puppet Class is a named block of Puppet code used to group related resources and configurations.

50. What is a Puppet Function?

Answer: A Puppet Function is a reusable piece of code that performs a specific operation, such as processing data or evaluating values within Puppet code.

QUICK PRACTICAL COMMANDS – REVISION

Tool	Command	Purpose
Git	git init	Initialize repository
Git	git status	Check file status
Git	git add .	Stage all changes
Git	git commit -m "message"	Save changes
Git	git push	Upload commits
Jenkins	Build Now	Execute a build
Docker	docker pull nginx	Download Nginx image
Docker	docker run -d -p 8080:80 nginx	Run Nginx container
Docker	docker ps	List running containers
Kubernetes	kubectl get pods	List Pods
Kubernetes	kubectl get nodes	List Nodes
Puppet	puppet --version	Check Puppet version
Puppet	puppet apply file.pp	Apply a Puppet manifest

END OF VIVA VOCE QUESTION BANK